

D879

Genetics

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A breeder with a prize-winning black guinea pig, bought for \$250, wants to know its genotype. Black (B) is dominant to white (b). She crosses it with a white guinea pig, and the litters produce 5 black and 4 white pups. What is the black parent's genotype?

- A. Bb
- B. BB
- C. bb
- D. It cannot be known, because the ratio is not exactly 1:1.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2. [TRANSFER — reversed stem / different variable]

An animal with the dominant phenotype is homozygous dominant (AA). What offspring are expected from a test cross?

- A. 1 dominant : 1 recessive
- B. All show the recessive phenotype.
- C. All show the dominant phenotype.
- D. 3 dominant : 1 recessive

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

In a dog breed popular with invented rescue groups, B (black) is dominant to b (brown), but a second gene controls pigment deposit: ee dogs are yellow no matter what B alleles they carry. Two BbEe dogs, both from the same kennel, are mated. What fraction of puppies are expected to be yellow?

- A. 1/16
- B. 4/16 (1/4)
- C. 9/16
- D. 3/16

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

A couple visiting an invented fertility clinic with a five-star rating are type A ($I^A i$) and type B ($I^B i$). What is the probability that their child has type O blood?

- A. 0, because neither parent is type O
- B. $3/4$
- C. $1/4$
- D. $1/2$

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5. [TRANSFER — reversed stem / different variable]

A child has type O blood and the mother is type A. Which father blood type is ruled out?

- A. Type A
- B. Type B
- C. Type O
- D. Type AB

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

A pedigree drawn on a whiteboard during a unit that runs through May shows two unaffected parents with three children, one of whom has a rare condition. The family has lived in the same town for four generations. What is the most likely mode of inheritance?

- A. Autosomal recessive
- B. Dominant, because it appears in the family
- C. Y-linked
- D. Autosomal dominant

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

E. coli grown in a flask on an orbital shaker at 37 °C are given lactose as their only sugar; no glucose is present. What is the state of the lac operon?

- A. Low, because glucose is absent
- B. Off, because the repressor is always bound
- C. Off, because lactose binds the operator and blocks it
- D. Transcribed at a high level

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8. [TRANSFER — reversed stem / different variable]

The lac operon is being transcribed at a high level. What must be true about the cell's sugars?

- A.** Both sugars are present.
- B.** Glucose is present and lactose is absent.
- C.** Lactose is present and glucose is absent.
- D.** Lactose is present; glucose does not matter.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

Identical twins raised in different cities have the same DNA sequence for a tumor-suppressor gene. One twin, who works night shifts, has heavy methylation at that gene's promoter, and the gene is barely expressed. What explains the difference?

- A.** An epigenetic change: promoter methylation silenced the gene.
- B.** Methylation turned the gene up.
- C.** A mutation changed the gene's sequence.
- D.** The twins inherited different alleles.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

A lab using a thermal cycler that cost \$8,000 starts PCR with 100 copies of a target sequence. The run takes about an hour. Assuming perfect doubling, how many copies are present after 10 cycles?

- A.** 1,000
- B.** 102,400
- C.** 1,024
- D.** 2,000

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11. [TRANSFER — reversed stem / different variable]

PCR starts with 50 copies and ends with 51,200 copies, assuming perfect doubling. How many cycles ran?

- A.** 512
- B.** 10
- C.** 20
- D.** 1,024

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

DNA fragments of 500, 2,000, and 5,000 base pairs are loaded with a blue tracking dye into a gel that ran for 45 minutes. Which fragment travels farthest from the wells?

- A.** The 5,000 bp fragment
- B.** Whichever fragment the blue dye marks
- C.** All travel the same distance, since all DNA has the same charge per length
- D.** The 500 bp fragment

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer A Concept 1

A breeder with a ~~prize-winning~~ black guinea pig, ~~bought for \$250~~, wants to know its genotype. Black (B) is dominant to white (b). She crosses it with a white guinea pig, and the litters produce 5 black and 4 white pups. What is the black parent's genotype?

Target word: genotype from test-cross results **Decides:** White pups appeared, so the black parent carried b.

A	Correct	A bb pup needs a b from each parent; about 1:1 fits Bb × bb.
B	Inverted operation	BB × bb gives all black pups.
C	Confusion set	A bb parent would be white.
D	Absolute language	Small litters vary; any white pup shows the parent is Bb.

2. Answer C Concept 1 • transfer

An animal with the dominant phenotype is homozygous dominant (AA). What offspring are expected from a test cross?

Target word: predict test-cross outcome **Decides:** AA × aa → all Aa.

A	Inverted operation	That is the result for a heterozygous parent.
B	Adjacent category	Every offspring gets a dominant A.
C	Correct	Every offspring is Aa.
D	Confusion set	That is an Aa × Aa cross.

3. Answer B Concept 2

In a dog breed ~~popular with invented rescue groups~~, B (black) is dominant to b (brown), but a second gene controls pigment deposit: ee dogs are yellow no matter what B alleles they carry. Two BbEe dogs, ~~both from the same kennel~~, are mated. What fraction of puppies are expected to be yellow?

Target word: fraction yellow **Decides:** ee occurs in 1/4 of offspring regardless of B.

A	Dropped element	Counts only bbee and misses BBee and Bbee.
B	Correct	Ratio is 9 black : 3 brown : 4 yellow.
C	Inverted operation	That is the black fraction.
D	Confusion set	Forces the cross into 9:3:3:1.

4. Answer C Concept 3

A couple ~~visiting an invented fertility clinic with a five-star rating~~ are type A (I^Ai) and type B (I^Bi). What is the probability that their child has type O blood?

Target word: probability of type O **Decides:** Each parent passes i half the time.

A	Absolute language	Both carry i and can each pass it on.
B	Inverted operation	That is the chance the child is not type O.
C	Correct	$\frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$ for ii.
D	Dropped element	Counts only one parent's chance of passing i.

5. Answer D Concept 3 • transfer

A child has type O blood and the mother is type A. Which father blood type is ruled out?

Target word: exclude a father **Decides:** The child is ii, so the father must carry i.

A	Confusion set	An I ^A i father can pass i.
B	Adjacent category	An I ^B i father can pass i.
C	Inverted operation	An ii father always passes i.
D	Correct	I ^A I ^B has no i to pass on.

6. Answer A Concept 4

A pedigree ~~drawn on a whiteboard during a unit that runs through May~~ shows two unaffected parents with three children, one of whom has a rare condition. ~~The family has lived in the same town for four generations.~~ What is the most likely mode of inheritance?

Target word: inheritance pattern **Decides:** An affected child of two unaffected parents.

A	Correct	Both parents must be carriers.
B	Confusion set	Showing up in a family does not make a trait dominant.
C	Adjacent category	Y-linked traits pass from affected father to all sons.
D	Inverted operation	A dominant trait would show in at least one parent.

7. Answer D Concept 5

E. coli grown ~~in a flask on an orbital shaker at 37 °C~~ are given lactose as their only sugar; no glucose is present. What is the state of the lac operon?

Target word: operon activity **Decides:** Lactose removes the repressor; no glucose means CAP is active.

A	Inverted operation	Glucose absence raises cAMP and increases transcription.
B	Absolute language	Lactose (allolactose) releases the repressor.
C	Confusion set	Lactose binds the repressor, not the operator.
D	Correct	Repressor off the operator and CAP-cAMP boosting RNA polymerase.

8. Answer C Concept 5 • transfer

The lac operon is being transcribed at a high level. What must be true about the cell's sugars?

Target word: work back from the result **Decides:** High output needs lactose present and glucose absent.

A	Adjacent category	Glucose keeps expression low through catabolite repression.
B	Inverted operation	That gives the lowest expression.
C	Correct	Both switches must be in the "on" position.
D	Dropped element	Leaves out CAP-cAMP control by glucose.

9. Answer A Concept 6

Identical twins ~~raised in different cities~~ have the same DNA sequence for a tumor-suppressor gene. One twin, ~~who works night shifts~~, has heavy methylation at that gene's promoter, and the gene is barely expressed. What explains the difference?

Target word: cause of different expression **Decides:** Promoter methylation silences genes without changing sequence.

A	Correct	Methylation blocks transcription while leaving the sequence the same.
B	Inverted operation	Promoter methylation usually silences genes.
C	Adjacent category	The stem says the sequences match.
D	Confusion set	Identical twins share alleles.

10. Answer B Concept 7

A lab ~~using a thermal cycler that cost \$8,000~~ starts PCR with 100 copies of a target sequence. ~~The run takes about an hour.~~ Assuming perfect doubling, how many copies are present after 10 cycles?

Target word: copies after n cycles **Decides:** 100×2^{10} .

A	Confusion set	Multiplies by the cycle count.
B	Correct	$100 \times 1,024 = 102,400$.
C	Dropped element	Leaves out the 100 starting copies.
D	Inverted operation	Multiplies by 2×10 instead of 2^{10} .

11. Answer B Concept 7 • transfer

PCR starts with 50 copies and ends with 51,200 copies, assuming perfect doubling. How many cycles ran?

Target word: solve for cycles **Decides:** $51,200 / 50 = 1,024 = 2^{10}$.

A	Inverted operation	Halves the fold change instead of finding the power of 2.
B	Correct	$2^{10} = 1,024$.
C	Confusion set	2^{20} would be about a million-fold.
D	Dropped element	Stops at the fold change and never takes the exponent.

12. Answer D Concept 8

DNA fragments of 500, 2,000, and 5,000 base pairs are loaded ~~with a blue tracking dye~~ into a gel ~~that ran for 45 minutes~~. Which fragment travels farthest from the wells?

Target word: fastest-moving fragment **Decides:** Smaller fragments move through the gel more easily.

A	Inverted operation	Largest fragments move slowest.
B	Camouflage	The dye tracks progress; it does not decide fragment movement.
C	Absolute language	Same charge per length, but the gel slows larger fragments more.
D	Correct	Smallest fragments slip through the gel fastest.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owled.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	5	
Wrong stem number	A figure copied from the prompt that is not the answer.	0	
Adjacent category	The right idea in the wrong bucket.	5	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	12	
Absolute language	Always, never, guarantee — where judgment applies.	4	
Confusion set	Every option is real; the test is exact role-matching.	9	
Camouflage	Answers a stem fact that was never the question.	1	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	The test cross	Cross a dominant-phenotype individual with a homozygous recessive; any recessive offspring prove the unknown is heterozygous, while all-dominant offspring point to homozygous dominant.	Expecting a 3:1 ratio from a test cross.	An unknown dominant individual crossed with a recessive one.
2	Epistasis	When one gene masks another, dihybrid ratios shift from 9:3:3:1 to patterns such as 9:3:4 (recessive epistasis).	Forcing every dihybrid cross into 9:3:3:1.	One gene controls whether another gene's trait shows at all.
3	Multiple alleles: ABO	ABO has three alleles: I^A and I^B are codominant and both dominant over i ; type O is ii .	Thinking two non-O parents cannot have an O child, or treating AB as blended.	Blood types of parents and children.
4	Reading pedigrees	Two unaffected parents with an affected child means the trait is recessive; an affected child always having an affected parent suggests dominant.	Calling a trait dominant because it appears often.	An affected child of unaffected parents.
5	lac operon regulation	The lac operon is transcribed strongly only when lactose is present (repressor removed) and glucose is absent (CAP-cAMP active).	Thinking lactose binds the operator, or that glucose switches the operon on.	Lactose and glucose levels in a bacterial culture.
6	Epigenetic silencing	DNA methylation at a promoter silences a gene without changing its sequence, and the pattern can pass through cell divisions.	Treating methylation as a mutation or as a gene activator.	Same DNA sequence, different gene expression.
7	PCR doubling	Each PCR cycle ideally doubles the target: copies = starting copies $\times 2^n$.	Multiplying by $2 \times n$ or forgetting the starting copies.	Starting copies and a number of cycles.
8	Gel electrophoresis	DNA is negatively charged and moves toward the positive electrode; smaller fragments travel farther.	Thinking larger fragments move faster.	Fragment sizes and a question about band position.

Course scope (public description): Genetics is an introductory course for undergraduate students seeking licensure or endorsement in secondary or middle grade science education. This course addresses the basic principles of heredity and the function of molecular genetics. Topics include Mendelian and non- Mendelian inheritance, gene regulation, and variation at the individual level. The course examines modern technologies and applications of genetics.